



MAI648 – HCI & Intelligent User Interfaces

PacketFlow: Visualizing the Invisible

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Abstract — This report details the conceptual design, architecture, and user-centered analysis of **PacketFlow**, an intelligent real-time network monitoring ecosystem. By bridging the gap between raw packet capture and human cognition, PacketFlow transforms high-speed telemetry into an interactive physics-based visualization, powered by local LLM reasoning and adaptive interfaces.

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1 Idea and Conceptual Design: PacketFlow

1.1 The Concept: What is PacketFlow?

PacketFlow is an intelligent, real-time network monitoring ecosystem designed to *visualize the invisible*. It bridges the gap between raw packet capture and human cognition by transforming high-speed network telemetry into an interactive, visual, and conversational interface.

Unlike traditional tools that present static lists or grouped abstractions, PacketFlow offers a granular, physics-based visualization of the network. It treats every IP address as a distinct node in a living ecosystem, allowing analysts to see exactly who is communicating with whom in real time, whether observing a live production network or simulating complex attack scenarios.

PacketFlow integrates local LLM-powered explanations, conversational querying, and an adaptive interface that adjusts to the user's expertise level (novice, intermediate, expert). The system incorporates proactive suggestions, glossary support, and contextual tooltips to reduce cognitive load, making the interface usable by both beginners and experienced analysts. Users can investigate events conversationally, receive real-time recommendations, and gain insight into anomalies without needing deep packet-analysis knowledge.

In addition to real-time monitoring, PacketFlow supports structured incident management, enabling users to group events, track status, and document investigations. These capabilities elevate the system beyond visualization and transform PacketFlow into an intelligent assistant for network defense.

1.2 The Architecture: A “Pure Display” Philosophy

PacketFlow follows a clean, separated architecture where the backend acts as the source of truth and the frontend serves as a pure visualization and interaction layer.

The Eye (Backend) A Python-based engine using TShark to capture real packets or generate complex mock scenarios. It handles:

- Multi-method anomaly detection (Z-score, IQR, EWMA, rate-based, protocol heuristics, etc.).
- Flow condensation and severity scoring.
- Local LLM reasoning (Ollama) for natural-language explanations.
- Event feedback processing (true/false positive signals).
- PCAP replay and mock-mode traffic generation.
- Structured event broadcasting via WebSocket.

The Lens (Frontend) A React/TypeScript application that receives high-frequency updates. It focuses on rendering performance and user interaction by providing:

- Smooth D3.js topology animations.
- Real-time event stream.
- AI chat interface for conversational investigation.
- Incident management module.
- Alert configuration panel.
- Adaptive UI modes (novice/intermediate/expert).
- Glossary and contextual help.
- Proactive suggestions.
- Keyboard shortcuts for rapid navigation.

This architectural separation ensures the interface remains responsive even under heavy load, while the backend manages intelligence and detection logic.

1.3 Addressing User Needs

PacketFlow addresses key operational pain points through its distinct features.

A. The Need for Granular Visibility

- ✗ **User Pain:** Traditional maps group IPs into generic “Internal” vs. “External” clouds, hiding lateral movement or specific compromised hosts.
- ✓ **Solution:** Individual IP addresses are rendered as distinct nodes (e.g., 192.168.1.105 → 8.8.8.8), enabling precise identification of suspicious activity.

B. The Need for Protocol Clarity

- ✗ **User Pain:** “TCP Traffic” is too vague; analysts need application-level detail.
- ✓ **Solution:** The Statistics Dashboard separates Transport and Application Layer traffic, highlighting DNS, HTTP, HTTPS, SSH, etc.

C. The Need for Safe Simulation & Training

- ✗ **User Pain:** Training typically requires unsafe real-world attack traffic.
- ✓ **Solution:** Mock Mode generates realistic synthetic scenarios, and PCAP replay enables reproducible academic or training exercises.

D. The Need for Immediate Awareness

- ✗ **User Pain:** Critical anomalies get buried in scrolling logs.
- ✓ **Solution:** Toast Notifications surface severe events instantly.

E. The Need for Explainability and Guidance

- ✗ **User Pain:** Traditional tools do not explain alerts.
- ✓ **Solution:**
 - Local LLM-generated explanations
 - Proactive suggestions
 - Glossary and tooltips for novice users

F. The Need for Structured Investigation

- ✗ **User Pain:** Analysts must juggle multiple systems to track incidents.
- ✓ **Solution:** Integrated incident management with notes, tags, event linking, and status tracking.

1.4 Operational Modes

Mode 1: Production (Mock Mode OFF)

- **Data Source:** Real network traffic (TShark).
- **Use Case:** Live SOC monitoring.
- **Features:** Anomaly detection, topology graph, event stream, AI chat, incident management, suggestions, adaptive UI.
- **Value:** Provides an accurate, real-time map of the environment.

Mode 2: Simulation (Mock Mode ON)

- **Data Source:** Backend simulation engine.
- **Use Case:** Training, demos, research.
- **Features:** PCAP replay, synthetic attack sequences, glossary support, novice mode.
- **Value:** Safe, repeatable, controlled attack simulations.

1.5 Visual Intelligence

The user interface is designed for immediate cognitive understanding:

- **Topology Coding:** Node colors represent type and anomaly severity.
- **Readability:** Monospace fonts highlight critical data (IP addresses, protocol names).
- **Focus:** Filters allow isolation of suspicious flows or high-severity anomalies.
- **Interactive Workflow:** Clicking nodes or edges enables filtering, AI explanations, and incident actions.
- **Adaptive Elements:**
 - Novice mode provides more explanations.
 - Expert mode prioritizes information density.

2 Target Groups

The primary target groups for the PacketFlow system were identified based on its capabilities, intended usage contexts, and the needs revealed during the analysis. These groups represent individuals who interact with network security tools across varying levels of expertise, operational demands, and educational or research environments.

Table 1: Summary of Target User Groups and Key Needs

User Group	Key Need from PacketFlow
Students	Natural-language explanations & Glossary
SOC Analysts	Real-time streaming & Incident Management
Net Admins	Actionable insights & Threshold configuration
Researchers	Privacy-preserving local AI & Mock Mode
Educators	PCAP Replay & Predictable learning environments
Power Users	Expert mode, shortcuts & Granular configuration

2.1 Cybersecurity Students and Trainees

These novice users engage with PacketFlow in academic labs, introductory cybersecurity courses, or training environments. Their goals include learning how network traffic behaves, understanding how anomalies are detected, and practicing incident investigation.

PacketFlow supports these needs through:

- natural-language explanations,
- glossary integration,
- PCAP replay mode,
- proactive suggestions,
- novice-friendly, low-cognitive-load interfaces.

Students benefit significantly from contextual help and guided workflows that scaffold their learning process.

2.2 SOC Analysts and Security Operators

These intermediate users require efficient real-time monitoring, triage, and incident tracking. PacketFlow provides:

- real-time event streaming,
- LLM-generated explanations,
- incident management features,
- robust filtering and search tools,
- keyboard shortcuts for rapid navigation.

They depend on accuracy, clarity, and actionable insights to maintain situational awareness during continuous operations.

2.3 Network Administrators and IT Personnel

These users monitor the health and behavior of organizational infrastructure. They require tools that offer:

- actionable network insights,
- customizable alert thresholds,
- IP allow/deny lists,
- clear anomaly diagnostics.

PacketFlow helps them quickly identify performance anomalies, configuration issues, or suspicious host behavior through its topology graph and flow analysis tools.

2.4 Cybersecurity Researchers and Academics

These expert users utilize PacketFlow for experimentation, evaluating detection models, or conducting studies in human–AI interaction. They require:

- privacy-preserving, local AI processing,
- PCAP replay,
- mock-mode simulation,
- expert-level detail views,
- configurable alert rules.

The system’s transparency and configurability support research on anomaly detection, adaptive interfaces, and user modeling.

2.5 Educators and Instructors

Educators use PacketFlow to support teaching activities, demonstrations, and structured exercises in network forensics or cybersecurity fundamentals. Features such as:

- PCAP replay,
- mock mode,
- guided explanations,
- adaptive UI modes,

enable instructors to design predictable learning environments and help students transition from novice to intermediate skill levels.

2.6 Advanced or Power Users

These experienced users—typically senior analysts or technically proficient researchers—benefit from:

- expert mode,
- high-velocity navigation using keyboard shortcuts,
- granular configuration of thresholds and rules.

They require minimal interface friction and the ability to rapidly explore flow metrics or experiment with rule-based alert conditions.

3 Needs and Verification Analysis

The contemporary digital landscape is defined by a paradox of visibility: while network telemetry data is generated at unprecedented volumes, the ability of human operators to synthesize this data into actionable intelligence is rapidly diminishing. This “Visibility Gap” represents a critical vulnerability in modern Security Operations Centers (SOCs), where the velocity of high-speed network traffic often outpaces human cognitive processing capabilities.

The proposed PacketFlow system addresses this disconnect by pioneering a “Pure Display” philosophy. It bridges raw machine telemetry and human cognition through physics-based, granular visualization. This analysis evaluates the theoretical foundations of PacketFlow, drawing from Information Visualization (InfoVis), Human-Computer Interaction (HCI), and Cognitive Psychology.

3.1 The Cognitive Science of Network Visualization

3.1.1 The Crisis of Abstraction and the “Hairball” Phenomenon

Visualizing network data is inherently challenging due to its complexity and high dimensionality. Traditional textual representations sever spatial and temporal relationships, forcing analysts into cognitively taxing mental reconstruction tasks.

Node-link diagrams, while intuitive, often suffer from the “hairball” phenomenon when edge density becomes high. Research shows that performance degrades significantly in graphs with more than 50–100 nodes unless advanced layout algorithms are applied [16]. Adjacency matrices avoid clutter but perform poorly for path-finding tasks necessary in cyber defense [12].

PacketFlow employs physics-based, force-directed layouts (e.g., Fruchterman-Reingold) to mitigate these challenges. Nodes repel each other, edges act as springs, and clusters form organically. This offloads grouping tasks to the visual cortex, enabling rapid recognition of communication patterns [14].

3.1.2 Granularity vs. Aggregation: The Node-Link-Group Debate

PacketFlow emphasizes granular representation of individual IP nodes. Research on node-link-group (NLG) diagrams shows that grouping aids high-level tasks but can obscure anomalies occurring on individual nodes [13]. Cybersecurity requires detecting low-frequency, high-impact outlier behavior.

Users also report higher engagement with NLG-style visuals [3], which is relevant given SOC fatigue risks. PacketFlow strikes a balance by letting clustering emerge organically from physics simulation rather than artificial grouping.

3.1.3 The Role of Animation and Temporal Dynamics

Cyber attacks unfold temporally. Static snapshots contribute to change blindness [15]. PacketFlow adopts animated transitions that preserve the analyst’s mental map. Smooth motion prevents disorientation when nodes shift positions [10].

3.1.4 Preattentive Processing Through Motion

Motion is processed preattentively by the visual system. PacketFlow’s animated particle flows along edges capitalize on this mechanism. Research shows particle motion improves accuracy in identifying flow direction and anomalies [8]. Excessive motion, however, can create noise, making filtering essential [18].

3.1.5 Cognitive Load and the Context-Switching Tax

SOC analysts suffer heavy cognitive load due to fragmented toolchains. The NASA-TLX instrument consistently shows high mental and temporal demand across traditional workflows [7]. PacketFlow reduces context switching by converting raw traffic into a unified visual representation, enabling perceptual “chunking” of patterns [9].

3.1.6 Dimensionality: The 2D vs. 3D Debate

While 3D and mixed reality (MR) visualizations can enhance spatial memory and team communication [1], they introduce occlusion and navigation friction for standard desktop use. PacketFlow uses optimized 2D force-directed layouts while leaving room for future “2.5D” enhancements (parallax cues, depth shadows) [20].

3.2 Operational Necessity: Visualizing the Kill Chain

3.2.1 The Challenge of Lateral Movement Detection

Lateral movement is a graph-theoretic path problem. Attackers traverse nodes via legitimate protocols, making log-based detection difficult. In a node-link graph, lateral movement appears as visual chains or starburst scans. Research shows that graph visualization improves detection of anomalous cross-cluster communication [5].

3.2.2 Protocol Clarity: Visualizing the OSI Stack

Distinguishing OSI layers visually is essential. Layer 4 traffic may reflect volumetric behavior, while Layer 7 reveals the semantic meaning (DNS tunneling, C2 beaconing). PacketFlow supports protocol filtering aligned with the InfoVis principle: “Overview first, zoom and filter, then details on demand” [21].

3.2.3 The “Toast” Dilemma: Designing Effective Notifications

Toast notifications must balance salience against interruption. Research recommends:

- contextual placement near affected nodes,
- consistent severity color coding,
- optimal display duration of 3–5 seconds with pinning options.

3.2.4 Automation Bias and Human-on-the-Loop

Automation bias leads analysts to trust system outputs blindly. PacketFlow counters this by showing raw traffic visually alongside alerts, keeping humans “on the loop” and capable of validating anomalies perceptually.

3.3 Simulation and Synthetic Environments

3.3.1 The Training Gap

Traditional training uses static logs, failing to teach temporal detection skills. PacketFlow’s Simulation Mode creates a high-fidelity cyber range environment [6], enabling realistic practice under time pressure.

3.3.2 Synthetic Traffic Generation

High-quality background noise is essential for credible training. PacketFlow can integrate GAN-based traffic generation as the field advances [2].

3.3.3 Developing Intuition Through Visual Repetition

Repeated exposure to visual attack signatures fosters instinctive recognition, consistent with perceptual learning literature [17].

3.4 Comparative Technology Review and Architecture

Table 2: Comparison of Network Analysis and Visualization Tools

Feature	Wireshark	EtherApe	Gephi	PacketFlow
Philosophy	DPI	Monitoring	Graph Analytics	Physics-Based
Representation	Text / Hex	Radial Graphs	Static Graph	Living Topology
Real-Time	Yes	Yes	No	Yes (WebSockets)
Granularity	Packet-Level	Protocol-Level	Metric-Level	Node + Flow
Target User	Forensic Analyst	Net Admin	Data Scientist	SOC Analyst
Architecture	Desktop App	Desktop App	Desktop App	Web App

3.5 Architectural Advantages

PacketFlow’s “Eye” (backend) and “Lens” (frontend) architecture provides:

- performance separation,
- browser accessibility,
- scalability across distributed sensors.

3.6 Verification Analysis Framework

3.6.1 User-Centered Design Methodology

PacketFlow adheres to User-Centered Design (UCD) principles [4]. Verification uses:

- System Usability Scale (SUS) [19]
- NASA-TLX workload measurements [11]

3.6.2 Stakeholder Questionnaire

The following survey was distributed to target user groups to validate requirements and gather qualitative data on current pain points.

1. What best describes your current role or status?

- Undergraduate / Graduate student
- Academic / Researcher
- Security Analyst / SOC practitioner
- Network / Systems Engineer
- IT Support / Administrator

- Software Developer
 - Cybersecurity trainee
 - Other
- 2. How often do you use network monitoring, security, or incident response tools (e.g., Wireshark, SIEM, IDS/IPS)?**
- Daily
 - Several times per week
 - Several times per month
 - A few times per year
 - Never
- 3. In what context do you mainly use (or expect to use) such tools?**
- Academic coursework, labs, or research projects
 - Professional SOC / security operations
 - Network administration / IT operations
 - Personal / home lab
 - I do not currently use such tools but may in the future
- 4. Which tools or platforms do you currently use or are familiar with?**
- Wireshark / tshark
 - SIEM platforms (e.g., Splunk, ELK, QRadar)
 - Intrusion Detection / Prevention Systems (IDS/IPS)
 - Firewalls and security gateways
 - Cloud security dashboards (e.g., AWS, Azure, GCP)
 - No specific tools
- 5. From your recent experience, what are the most important problems, limitations, or frustrations you have faced?**
(Open-ended response)
- 6. How familiar are you with AI-assisted or intelligent security tools?**
- Very familiar
 - Moderately familiar
 - Slightly familiar
 - Not familiar at all
- 7. Which capabilities would you consider most useful in an intelligent interface like PacketFlow?**
- Real-time anomaly detection and severity scoring
 - Natural language explanations of anomalies
 - Conversational querying (chat-like interface)
 - Adaptive interface (novice/intermediate/expert)

- Proactive suggestions for investigation
- Integrated incident management
- Network topology visualization
- Glossary and contextual help
- Strictly local processing (privacy-first)

8. Please rate how important each of the following characteristics is for an intelligent network security interface.

(Rated on a Likert Scale from “Not important” to “Extremely important”)

- Ease of learning and use
- Transparency and explainability of AI outputs
- Trust in anomaly detections and recommendations
- Real-time performance and responsiveness
- Ability to control or override AI suggestions
- Data privacy and local processing (no cloud)
- Support for incident management and collaboration
- Educational value (helping me learn security concepts)

9. Which types of support or guidance would you find most helpful?

- Step-by-step investigation suggestions
- Example queries for the AI component
- Contextual tooltips and definitions (glossary)
- Visual summaries and dashboards
- Keyboard shortcuts and efficiency features
- Tutorials or guided walkthroughs

10. What would be your main goals and expectations for a privacy-first interface like PacketFlow?

(Open-ended response)

11. Is there any functionality or AI assistance essential for such a system not mentioned so far?

(Open-ended response)

3.7 Survey Results and Analysis

A preliminary survey was conducted with a diverse group of participants ($N = 8$), ranging from Cybersecurity Trainees and Students to IT Managers and Security Researchers. The responses provided critical validation for PacketFlow’s core design philosophy.

3.7.1 Key Findings

- **Validation of the “Visibility Gap”:** Respondents cited significant frustrations with current tools, specifically noting that “tools overwhelm students” and complaining of “random alerts and limited visibility.” One respondent explicitly highlighted the need for “more clear and meaningful insights,” confirming the necessity of PacketFlow’s AI-driven explanation layer.
- **Support for Adaptive Interfaces:** The demographic spread—from daily users of network tools to those who use them “a few times per year”—validates the MoSCoW requirement for an **Adaptive Interface**. A “one-size-fits-all” dashboard would overwhelm the novices identified in the survey while frustrating the experts.
- **The Importance of Trust and Privacy:** When rating system characteristics, *Data Privacy* and *Trust in Anomaly Detections* received the highest volume of “Extremely Important” ratings. This strongly supports the architectural decision to use **Local LLMs (Ollama)** rather than cloud-based APIs, ensuring telemetry data never leaves the premises.
- **Demand for Explainability:** Participants ranked “Natural language explanations of anomalies” and “Real-time anomaly detection” as the most useful capabilities. This mirrors the frustration with legacy tools (e.g., Wireshark) where raw data is abundant but context is scarce.

3.7.2 Implications for Design

The survey results confirm that while raw telemetry is essential for experts, the primary barrier to entry is cognitive load. Consequently, the design priority of PacketFlow remains the **reduction of complexity** through visual abstraction and conversational assistance, rather than simply displaying more data rows.

4 Conclusion and Future Outlook

PacketFlow stands as a necessary evolution in security visualization. Its physics-based “Pure Display” architecture bridges human perceptual strengths and machine telemetry. As node counts scale in IoT and edge networks, PacketFlow’s granular, dynamic approach becomes essential.

Crucially, our stakeholder analysis validates this direction. The survey results highlighted a critical need for **data privacy** and **reduced cognitive load**, confirming that the “Visibility Gap” is a human problem as much as a technical one. By integrating local, privacy-preserving AI reasoning with an adaptive interface, PacketFlow successfully meets the diverse needs of students, analysts, and researchers alike.

Future iterations will focus on further enhancing these capabilities, integrating “2.5D” visual cues for deeper topological analysis and expanding the generative adversarial network (GAN) capabilities for even more realistic simulation training.

5 Personas and Scenarios

5.1 Persona 1 — “Elena Costa” (Novice Cybersecurity Student)

Profile Overview

Age:	21
Background:	Undergraduate cybersecurity student
Technical Expertise:	Beginner; basic Wireshark and Linux experience

Goals

- Understand network traffic fundamentals.
- Learn how anomalies and attacks are detected.
- Build confidence analyzing security events.
- Develop familiarity with real-world investigation workflows.

Frustrations

- Traditional tools display overwhelming packet-level detail.
- Difficult to interpret alerts without explanations.
- Uncertain where to begin when identifying suspicious traffic.

Motivations

- Prefers learning through natural-language explanations.
- Values interfaces that guide next steps.
- Appreciates visual aids and contextual help.

Scenario 1 — Learning and Exploration

Elena opens PacketFlow during a university lab session and loads a PCAP file using replay mode. The system immediately condenses traffic into flows and detects several DNS-related anomalies. Each anomaly appears in the event stream accompanied by a plain-language explanation generated by the LLM, helping her understand why a behavior is considered suspicious.

Curious about specific terms, she clicks highlighted glossary icons to access definitions such as “anomaly score” and “suspicious domain pattern.” The topology graph provides a simplified visualization of communication patterns, allowing her to clearly see which hosts are interacting.

When she opens one event, proactive suggestions appear—recommending that she filter traffic by the suspicious IP or compare the flow with historical behavior. She follows these suggestions to explore the investigation workflow.

The adaptive interface keeps explanations simple because she is in novice mode, showing essential details while allowing deeper insights on demand.

By completing the scenario, Elena gains confidence in identifying anomalies without feeling overwhelmed.

5.2 Persona 2 — “Ahmed Rahman” (Intermediate SOC Analyst)

Profile Overview

Age:	32
Background:	Works in a university SOC; handles alert triage
Technical Expertise:	Moderate; familiar with SIEMs, firewalls, and packet analysis

Goals

- Quickly assess whether an anomaly represents a real threat.
- Efficiently correlate related alerts.
- Maintain situational awareness across the network.

Frustrations

- High false-positive rates in legacy tools.
- Constant switching between dashboards slows investigations.
- Alerts often lack context or explanatory details.

Motivations

- Wants accurate, contextualized anomaly information.
- Values fast navigation, such as keyboard shortcuts.
- Prefers a unified interface for triage and incident tracking.

Scenario 2 — Real-Time Anomaly Investigation

During his SOC shift, Ahmed monitors PacketFlow’s real-time event stream, where several HIGH and CRITICAL anomalies appear. He selects a CRITICAL alert and immediately receives an LLM-generated explanation summarizing why the traffic is dangerous and suggesting potential next steps.

To organize his findings, Ahmed creates an incident in the incident management module, linking relevant anomaly events. The system transitions to the incident workspace while preserving his UI state. He adds notes, assigns tags, and marks the incident status as “investigating.”

As repetitive anomalies arise from a particular source, Ahmed receives proactive suggestions advising him to apply filters. Using keyboard shortcuts, he rapidly switches between the topology, event stream, and incident panels.

Finally, he marks one anomaly as a false positive, contributing feedback to improve detection reliability. The combined presence of explanations, visual context, and workflow tooling allows him to escalate the real threat more confidently and efficiently.

5.3 Persona 3 — “Dr. Maria Nikolaou” (Network Security Researcher)

Profile Overview	
Age:	45
Background:	Academic researcher focused on anomaly detection and privacy-preserving analytics
Technical Expertise:	Expert in machine learning and network security

Goals

- Experiment with detection models and traffic behavior.
- Evaluate AI-generated explanations and adaptive intelligent UI features.
- Conduct research using privacy-safe, locally processed data.

Frustrations

- Cloud-based tools conflict with privacy and compliance requirements.
- Many anomaly detection interfaces lack transparency and flexibility.
- Existing platforms rarely support experimentation or teaching workflows.

Motivations

- Prefers open-source, configurable tools.
- Values local LLM processing for sensitive data.
- Wants insights into both data and adaptive UI behavior.

Scenario 3 — Research and Model Experimentation

Maria uses PacketFlow in mock mode to simulate network scenarios for her research group. She alternates between synthetic streams and PCAP replay mode to compare how the system behaves on different datasets.

The event stream displays anomalies detected via eight statistical and heuristic methods, and she evaluates how each contributes to the severity score. Switching to the chat panel, she queries the LLM about observed patterns and receives detailed explanations referencing flows and protocols, providing a readable layer on top of the detection model metrics.

Maria tests the expert mode, revealing additional details such as per-method detection contributions, confidence indicators, and deeper technical terminology. She experiments with alert configuration options, adjusting thresholds, sensitivity, and IP allow/deny lists to observe how they affect anomaly detection.

Real-time visualization, local AI reasoning, and adaptive interface behavior enable her to conduct privacy-preserving research without external dependencies.

Summary: How Personas Inform Design

The specific needs of these three personas directly dictated the functional requirements and interface design of PacketFlow. The table below maps each user profile to their corresponding system features.

6 Main Tasks

6.1 Task 1: Investigate a Detected Network Anomaly

Short Description

The user selects an anomaly from the real-time event stream and reviews its AI-generated natural-language explanation to understand why it was flagged. The user inspects related flows, explores the network topology graph, applies filters or suggestions recommended by the system, and determines whether the behavior represents malicious activity. The user may also consult

Table 3: Mapping Personas to Design Features

Persona	Design Features & Requirements
Elena	• Educational explanations • Guided suggestions • Simplified UI mode • Glossary and contextual help • Low cognitive load • PCAP replay support for learning
Ahmed	• Real-time triage capabilities • Accurate and contextual anomaly explanations • Fast navigation (keyboard shortcuts) • Incident management workflow • Proactive recommendations • Feedback labeling to improve detection accuracy
Maria	• Privacy-first local AI processing • Configurable alert thresholds and rule sets • PCAP replay and mock mode for experimentation • Detailed anomaly metrics in expert mode • Transparency in detection and explanation logic • Flexible topology and event views

the glossary and contextual help, request deeper analysis through the conversational AI chat, and optionally label the event as a true or false positive to improve system accuracy.

Target User Group & Frequency

This task is primarily performed by SOC analysts and network administrators, who conduct anomaly investigations multiple times per day during continuous monitoring. Cybersecurity students perform this task during training labs or PCAP replay sessions, typically on a weekly basis. Researchers may perform extended analysis sessions during experimentation using mock mode or historical datasets.

6.2 Task 2: Create and Manage a Security Incident

Short Description

After identifying suspicious or correlated events, the user creates a new incident in the incident management module. The user links relevant events, adds notes or tags, assigns the incident to themselves or team members, and updates the status through the lifecycle stages (open → investigating → resolved/false positive). Users may attach additional flows, consult AI explanations for context, and leverage proactive suggestions that indicate possible incident

grouping or follow-up actions. Status changes and updates persist across the interface through the system's real-time state management.

Target User Group & Frequency

This task is mainly performed by SOC analysts and security researchers, who update and track multiple incidents daily. It is also relevant for advanced cybersecurity students who conduct structured investigations in academic environments, where incident management may be performed several times per semester or during simulation exercises. Researchers performing model evaluation may use incidents to categorize findings across datasets.

6.3 Task 3: Configure Detection Sensitivity and Custom Alert Rules

Short Description

The user adjusts the global anomaly sensitivity slider, modifies thresholds for metrics such as anomaly score or flow rate, manages IP whitelists and blacklists, and creates simple rule-based conditions that trigger actions (e.g., notifications or incident creation). Users may switch between novice and expert interface modes, enabling more detailed configuration options for advanced users. Configuration changes influence the detection pipeline in real time, allowing rapid experimentation during PCAP replay or mock mode sessions.

Target User Group & Frequency

This task is typically performed by network engineers, researchers, and advanced SOC analysts, who tune the system according to operational needs, dataset characteristics, or research objectives. Configuration is often performed initially during deployment and refined periodically (weekly or monthly) as network behavior evolves. Novice students may perform this task only during guided exercises, especially when exploring how thresholds affect anomaly detection outcomes.

7 Requirements (MoSCoW Analysis)

7.1 Must Have (The Core)

Real-Time Anomaly Detection and Flow Processing

Justification: Fundamental to PacketFlow's purpose as an intelligent network security monitoring system. SOC analysts, students, and researchers rely on real-time visibility and responsive anomaly identification for situational awareness.

Local, Privacy-Preserving AI Reasoning (Ollama / On-Device LLM)

Justification: Required to meet privacy, compliance, and research constraints. Personas such as Maria and Ahmed emphasize the need for full local processing to avoid exposing sensitive telemetry to external services.

Transparent AI Explanations of Alerts and Anomalies

Justification: Critical for trust formation, understanding system decisions, and reducing cognitive load, especially for novices. Explanations enable analysts to validate alerts rather than blindly trusting automated detection.

Incident Management Workflow (open → investigating → resolved/false positive)

Justification: Central to professional security operations and essential for structured investigations. Supports documentation, collaboration, and lifecycle tracking.

Core Visualizations: Real-Time Event Stream, Topology Graph, Flow and Packet Context

Justification: Users must quickly understand host interactions and behavior patterns. Visual situational awareness is crucial for novices and experts alike, supporting both learning and operational triage.

Conversational AI Query Interface

Justification: Enables natural language interaction with network data and explanations. Essential for novice usability and for expert users seeking quick reasoning, summaries, or clarifications.

7.2 Should Have (Important)

Adaptive Interface Based on User Expertise (novice/intermediate/expert)

Justification: Improves usability and supports varied skill levels. Novices benefit from simplified views, while experts gain access to richer detail and faster workflows.

Proactive Suggestions for Actions and Investigations

Justification: Helps users navigate complex data by surfacing relevant investigative actions, filters, or insights. Improves efficiency and reduces manual effort.

Feedback Labeling System (true/false positives)

Justification: Provides user-driven refinement of anomaly detection quality, creating a feedback loop for improving future accuracy and supporting user engagement.

Search and Filter Capabilities for Events and Incidents

Justification: Necessary for operation at scale, enabling users to focus on relevant subsets of data. Especially important during high-volume monitoring and historical investigations.

PCAP Replay and Mock Mode for Testing and Training

Justification: Supports controlled experimentation, research evaluation, and educational use. Allows testing without requiring a live network capture environment.

7.3 Could Have (Desirable)

Keyboard Shortcuts for Power Users

Justification: Improves workflow efficiency for experienced users. Not required for core functionality and can be introduced incrementally.

Glossary and Contextual Learning Module

Justification: Provides educational value for students and less experienced analysts. The system remains functional without it, but usability and learning outcomes are improved significantly.

Custom Rule Builder (if-condition → action)

Justification: Adds flexibility for advanced users who want automation or custom monitoring behavior. Not essential for baseline detection.

Fine-Grained Threshold Controls (anomaly score, packet rate, byte rate)

Justification: Useful for environments requiring tailored detection sensitivity. Experts benefit most, while novices may not require granular control.

Advanced Visual Enhancements (e.g., extended graph options, layout customization)

Justification: Enhances user experience but does not affect the core functionality of anomaly detection or incident management.

7.4 Won't Have (Out of Scope)

Cloud-Based Data Processing or External API Dependence

Justification: Contradicts the system's privacy-first design and violates security expectations from researchers and SOC analysts. All processing must remain fully local.

Automated Blocking, Firewall Manipulation, or Network Reconfiguration

Justification: Introduces high operational risk and exceeds the intended scope of an academic intelligent interface prototype. Could result in unintended disruptions.

Unrelated Functionalities (e.g., email alerting, generic chatbots, unrelated analytics)

Justification: Adds unnecessary complexity without contributing to the core focus of intelligent network monitoring or user-centered intelligence.

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